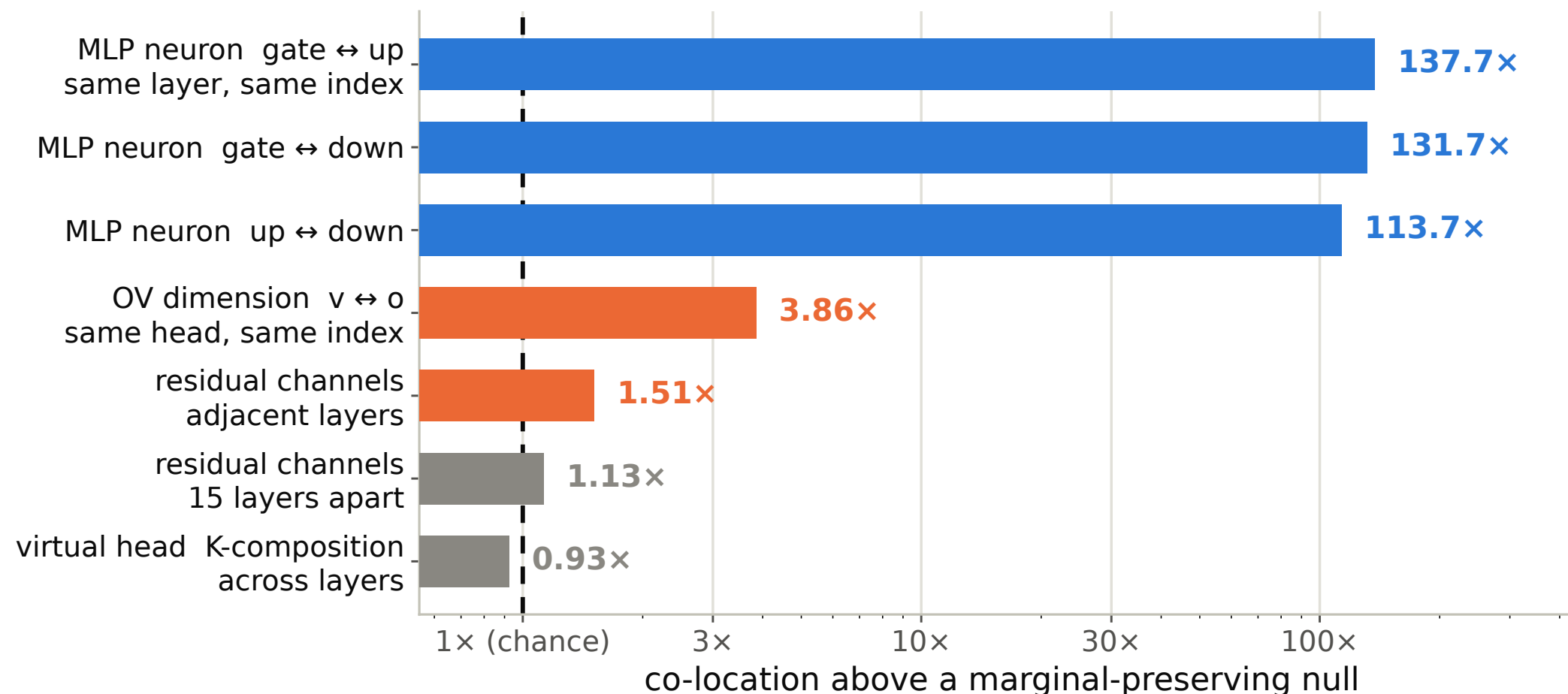


How do parameters inside one component interact?

Each bar is a co-location statistic against a null that preserves every marginal and destroys only the pairing, so they are comparable.

A component binds a computational UNIT almost perfectly — in SwiGLU, gate row i , up row i and down column i are one neuron, and components own them together at $138\times$ chance ($89\times$ above a layer-shifted control). Binding then decays: loosely within a head, barely across adjacent layers, and at exactly chance for the composed head pairs that make a virtual attention head.

Interaction falls off with structural distance



Cross-layer coupling is real but local

